导数

$y = x^{\mu}\ (\mu \in R)$

$$y' = \lim_{\Delta x \to 0} \frac{f(x_0+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{(x_0+\Delta x)^{\mu} - x_0^{\mu}}{\Delta x}$$

若 $x_0 = 0$，$\lim\limits_{\Delta x \to 0} \frac{(\Delta x)^{\mu}}{\Delta x} = \lim\limits_{\Delta x \to 0} (\Delta x)^{\mu-1} = \begin{cases} 0 & \mu > 1 \\ 1 & \mu = 1 \\ \text{不存在} & \mu < 1 \end{cases}$

若 $x_0 \neq 0$，$\lim\limits_{\Delta x \to 0} \dfrac{x_0^{\mu-1}\left(\left(1+\frac{\Delta x}{x_0}\right)^{\mu} - 1\right)}{\frac{\Delta x}{x_0}} = \mu x_0^{\mu-1}$

$\mu x_0^{\mu-1}$

设 $x = \varphi(y)$ 在区间 I 上单调，区间 I 内一点 y 处 $\varphi'(y)$ 存在不为零

则 $y = f(x)$ 在 $x = \varphi(y)$ 处也可导. 且 $f'(x) = \frac{1}{\varphi'(y)}$

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta y \to 0} \frac{1}{\frac{\Delta x}{\Delta y}} = \frac{1}{\varphi'(y)}$$

1. $y = \arctan x$，$x = \tan y$

$$y' = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}$$

2. $y = \arcsin x$，$x = \sin y$　　$x \in (-1,1)$ 时. $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$y' = \frac{1}{\cos y} = \frac{1}{\cos(\arcsin x)} = \frac{1}{\sqrt{1-x^2}}$$

对于一个反导数，原函数可能不只一个

但其值之差为常数

3 $y = \arccos x$，$x = \cos y$

$$y' = \frac{1}{-\sin y} = \frac{-1}{\sqrt{1-x^2}}$$

复合函数求导证明 $y=f(\varphi(x))$

$$\frac{dy}{dx}=f'(x)=\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x}=\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta u}\cdot\frac{\Delta u}{\Delta x}=f'(u_0)\cdot\varphi'(x_0)\quad(\Delta u\neq 0)$$

仅在 X 严格单调时成立.

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$$f'(u_0)=\lim_{\Delta u\to 0}\frac{f(u_0+\Delta u)-f(u_0)}{\Delta u}\Leftrightarrow\lim_{\Delta u\to 0}\frac{f(u_0+\Delta u)-f(u_0)-f'(u_0)\Delta u}{\Delta u}=0$$

$$w_{f,u_0}(\Delta u)=\begin{cases}\frac{f(u_0+\Delta u)-f(u_0)}{\Delta u}-f'(u_0), & \Delta u\neq 0\\ 0 & \Delta u=0\end{cases}$$

而 $\lim_{\Delta u\to 0} w_{f,u_0}(\Delta u)=0=w_{f,u_0}(0)$

$w_{f,u_0}(\Delta u)$ 在 $\Delta u=0$ 处连续

此时 $f(u_0+\Delta u)-f(u_0)=(f'(u_0)+w_{f,u_0}(\Delta u))\cdot\Delta u$

$\overset{\shortparallel}{\varphi(x_0)}$

$\Delta u=\varphi(x_0+x)-\varphi(x_0)$

得 $f(\varphi(x_0+\Delta x))-f(\varphi(x_0))=(f'(\varphi(x_0))+w_{f,\varphi(x_0)})(\varphi(x_0+\Delta x)-\varphi(x_0))$

令 $\Delta x\to 0$. $\lim_{\Delta x\to 0}(\varphi(x_0+\Delta x)-\varphi(x_0))=0$

隐函数求导

$$x\cdot y-e^x+e^y=0$$

视 $y=y(x)$ 为中间变量

$$y+x\cdot y'-e^x+e^y\cdot y'=0$$

$$y'=\frac{e^x-y}{x+e^y}$$

$$\frac{\partial F}{\partial x}=y-e^x \qquad \frac{\partial F}{\partial y}=x+e^y \Rightarrow \frac{dy}{dx}=-\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}=\frac{e^x-y}{x+e^y}$$

计算高阶导时，不要留下 y'

曲线 $x^3+3y^5-2xy=7$ 切点 $M(2,1)$ 的切线和法线方程

$$3x^2+15y^4\cdot y'-2y-2xy'=0$$

$$y'=\frac{2y-3x^2}{15y^4-2x} \quad \text{代入}(2,1)$$

$$\Rightarrow y'=-\frac{10}{11} \qquad 10x+11y-31=0$$

参数方程

$x=\varphi(t)$ ∃反函数 $t=\varphi^{-1}(x)$

$y=\psi(t)=\psi\circ\varphi^{-1}(x)$

$$\frac{dy}{dx}=\frac{dy}{dt}\cdot\frac{dt}{dx}=\frac{\psi'(t)}{\varphi'(t)}$$

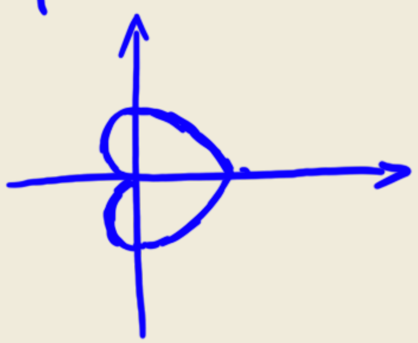
$$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{d}{dt}(y'_x)\cdot\frac{dt}{dx}$$

例 $x=a\cos t$, $y=b\sin t$ 求 $\frac{dy}{dx}$ 设 $t=\frac{\pi}{6}$

$$\frac{dy}{dx}=-\frac{b\cos t}{a\sin t} \qquad k=\frac{dy}{dx}\Big|_{t=\frac{\pi}{6}}=-\frac{\sqrt{3}b}{a}$$

极坐标

$\rho=1+\cos\theta$



$$\begin{cases} x=\rho(\theta)\cdot\cos\theta \\ y=\rho(\theta)\cdot\sin\theta \end{cases}$$

$$\frac{dy}{dx}=\frac{\rho'(\theta)\sin\theta+\rho(\theta)\cos\theta}{\rho'(\theta)\cos\theta-\rho(\theta)\sin\theta}$$

莱布尼茨公式

$$(uv)^{(n)}=\sum_{k=0}^{n}C_n^k u^{(n-k)}v^{(k)}=u^{(n)}v+C_n^1u^{(n-1)}v'+C_n^2u^{(n-2)}v''+\dots+C_n^ku^{(n-k)}v^{(k)}+\dots+uv^{(n)}$$

当 $n=1$ 时 $(uv)'=u'v+uv'$ 成立

若 n 时满足，考虑 $n+1$

$$(uv)^{n+1}=u^{n+1}v+u^nv'+C_n^1\cdot(u^nv'+u^{n-1}v'')+C_n^2(u^{(n-1)}v''+u^{(n-2)}v''')+\dots+u'v^n+uv^{n+1}$$

而 $C_n^{k-1}+C_n^k=C_{n+1}^k$

$$\Rightarrow (uv)^{n+1}=u^{n+1}v+C_{n+1}^1u^nv'+\dots+uv^{(n+1)}$$

$$=\sum_{k=0}^{n+1}C_{n+1}^k u^{(n+1-k)}v^{(k)}$$

设 $y=\arcsin x$ 求 $y^{(n)}(0)$

$$y'=\frac{1}{\sqrt{1-x^2}} \Leftrightarrow (1-x^2)(y')^2=1 \Rightarrow -2x(y')^2+2(1-x^2)y'\cdot y''=0$$

$$(1-x^2)y''=x\cdot y' \Rightarrow y''(0)=0$$

对上式关于 x 求 $n-2$ 阶导

$$(1-x^2)y^{n}+C_{n-2}^{1}(-2x)y^{(n-1)}+C_{n-2}^{2}(-2)\,y^{(n-2)}=x\cdot y^{(n-1)}+C_{n-2}^{1}\,y^{(n-2)}$$

令 $x=0$ 则 $y^{n}(0)-(n-2)(n-3)\,y^{(n-2)}(0)=(n-2)\,y^{(n-2)}(0)$

即 $y^{(n)}(0)=(n-2)^2\cdot y^{(n-2)}(0)$ $\quad n\geq 3$

$y'(0)=1 \quad y''(0)=0$

$$y^{n}(0)=\begin{cases}0 & n\text{ 为偶}\\ ((n-2)!!)^2 & n\text{ 为奇}\end{cases}$$

$(n-2)!!=(n-2)(n-4)\cdots\times 7\times 5\times 3\times 1$